Strict quadratic Lyapunov decay for a torus map

**Abstract**

Buescu and Oliveira proposed a piecewise quadratic Lyapunov function for a torus map arising from three coupled clocks. We verify strict decrease of this exact function for every parameter $0 < a < 1/3$. In circular-gap coordinates, the function is a centered sum of three squares. The update does not decrease the smallest gap or increase the largest gap, and it preserves their order. An elementary three-coordinate majorization argument then gives a one-step energy loss of at least half the squared gap displacement. The formulas agree across the torus seams, and the loss vanishes exactly at the six fixed points. This verifies a strict-decrease property of the proposed function; Conley completeness is a separate requirement.

**Keywords:** Discrete Lyapunov function; torus map; circular gaps; majorization

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# 1 Introduction and main result

Let $0 < a < 1/3$, and let $G_a : \mathbb{T}^2 \to \mathbb{T}^2$, where $\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$, be induced by the lift

$$\begin{aligned}\widetilde{G}_a(x,y) = \big(&x + a[2\sin x + \sin y + \sin(x-y)],\\ &y + a[\sin x + 2\sin y + \sin(y-x)]\big).\end{aligned} \tag{1}$$

The increments are $2\pi$-periodic in each variable, so the lift defines a torus map. Buescu and Oliveira [1] study this discrete model of three coupled clocks. Their Theorems 6 and 7 establish basinwise Lyapunov functions and attraction to two synchronized states. We consider the different quadratic function proposed in their Conjecture 1.

Put $c = 2\pi/3$. For representatives $(x, y) \in [0, 2\pi)^2$, the function in Conjecture 1 of [1] is

$$L(x,y) = \begin{cases}(x-c)^2 + (y-2c)^2 - (x-c)(y-2c), & y \geq x,\\ (y-c)^2 + (x-2c)^2 - (y-c)(x-2c), & y < x.\end{cases} \tag{2}$$

To express its decrement, define

$$\begin{gathered}\Delta = \{r \in \mathbb{R}^3 : r_i \geq 0,\ r_1 + r_2 + r_3 = 2\pi\},\\ F_i(r) = 3\sin r_i - \sum_{j=1}^{3} \sin r_j, \qquad T_a r = r + aF(r),\\ E(r) = \frac{1}{2}\sum_{i=1}^{3}(r_i - c)^2.\end{gathered} \tag{3}$$

The norm below is the ordinary Euclidean norm on $\mathbb{R}^3$.

**Theorem 1.** *For every fixed $0 < a < 1/3$, the formulas in (2) define a continuous nonnegative function on $\mathbb{T}^2$, with zero set*

$$\{p_+, p_-\}, \qquad p_+ = (c, 2c), \quad p_- = (2c, c).$$

*The lift $\widetilde{G}_a$ preserves both closed triangles*

$$A = \{0 \le x \le y \le 2\pi\}, \qquad B = \{0 \le y \le x \le 2\pi\}.$$

*On these triangles, respectively, the gap coordinates*

$$r = (x, y - x, 2\pi - y), \qquad r = (y, x - y, 2\pi - x) \tag{4}$$

*obey exactly the update $r^+ = T_a r$, and $L(x, y) = E(r)$. In particular, $T_a(\Delta) \subseteq \Delta$, and for every $r \in \Delta$,*

$$E(T_a r) \le E(r) - \frac{1}{2}\|T_a r - r\|^2. \tag{5}$$

*Consequently, for every $z \in \mathbb{T}^2$ and every compatible representation (4),*

$$L(G_a z) \le L(z) - \frac{1}{2}\|T_a r - r\|^2. \tag{6}$$

*The squared displacement in this formula is independent of the choice of a compatible closed-triangle representative. Moreover,*

$$\begin{aligned} L(G_a z) = L(z) \quad &\Longleftrightarrow \quad z \in \mathrm{Fix}(G_a), \\ \mathrm{Fix}(G_a) &= \{(0,0), (0,\pi), (\pi,0), (\pi,\pi), p_+, p_-\}. \end{aligned} \tag{7}$$

*Thus $L$ strictly decreases at every point outside these six fixed points.*

The formula (2) is due to Buescu and Oliveira. Our proof writes out the classical three-coordinate majorization argument [2], including the quadratic remainder. The estimate concerns a finite discrete update, not the derivative of an energy along a continuous-time system.

# 2 The torus function and its gap representation

We first check that the two branches of (2) fit the quotient. Their polynomial extensions to $A$ and $B$ are

$$\begin{aligned} Q_A(x, y) &= x^2 + y^2 - xy - 2\pi y + \frac{4\pi^2}{3}, \\ Q_B(x, y) &= x^2 + y^2 - xy - 2\pi x + \frac{4\pi^2}{3}. \end{aligned} \tag{8}$$

They agree when $x = y$. For $0 \le t \le 2\pi$, direct substitution gives

$$\begin{aligned} Q_A(0, t) = Q_B(2\pi, t) &= t^2 - 2\pi t + \frac{4\pi^2}{3}, \\ Q_B(t, 0) = Q_A(t, 2\pi) &= t^2 - 2\pi t + \frac{4\pi^2}{3}. \end{aligned} \tag{9}$$

The common diagonal value is the same polynomial in $t$. These identities include the four corners. Hence the branches glue to a continuous function on the closed square which is constant under its edge identifications. This is precisely a continuous function on $\mathbb{T}^2$.

For $(x, y) \in A$, set $u = x - c$ and $v = y - 2c$. The first representation in (4) gives

$$r - c\mathbf{1} = (u, v - u, -v),$$

and therefore

$$E(r) = \frac{1}{2}\big[u^2 + (v-u)^2 + v^2\big] = u^2 + v^2 - uv = Q_A(x, y). \tag{10}$$

The identity on $B$ follows by exchanging $x$ and $y$. In particular, $L \geq 0$, and equality holds only at $p_+$ in $A$ or $p_-$ in $B$.

Write $(X, Y) = \widetilde{G}_a(x, y)$. On $A$ we have

$$\sin r_1 = \sin x, \qquad \sin r_2 = \sin(y - x), \qquad \sin r_3 = -\sin y.$$

Substitution into the literal lift, before any reduction modulo $2\pi$, gives

$$(X, Y - X, 2\pi - Y) = r + aF(r) = T_a r. \tag{11}$$

For example, the first increment is $a(2\sin r_1 - \sin r_2 - \sin r_3)$, and the second is $a(-\sin r_1 + 2\sin r_2 - \sin r_3)$. The third follows by summing the gaps to $2\pi$. The map (1) commutes with exchanging $x$ and $y$, so on $B$ the corresponding identity is

$$(Y, X - Y, 2\pi - X) = T_a r. \tag{12}$$

No strict inequality or division was used, so these identities hold on every edge and on the diagonal. To conclude that the updated gaps are nonnegative, we next prove invariance of $\Delta$ directly.

# 3 The extremal gaps and preservation of order

The vector field $F$ in (3) commutes with every permutation of the three coordinates, and $\sum_i F_i(r) = 0$. Thus we may sort an arbitrary gap vector as

$$0 \leq m \leq b \leq M, \qquad m + b + M = 2\pi.$$

Here $F_m = 2\sin m - \sin b - \sin M$ and $F_M = 2\sin M - \sin m - \sin b$ denote the components at the smallest and largest gaps.

**Lemma 1.** *For every such triple,*

$$F_m \geq 0, \qquad F_M \leq 0.$$

*Moreover, $F_m > 0$ whenever $0 < m < c$.*

**Proof.** The sum condition gives $m \le c \le M$. By sine addition,

$$F_m = 2\sin(m/2)\big[2\cos(m/2) - \cos((M-b)/2)\big]. \tag{13}$$

Since $0 \le m/2 \le \pi/3$, we have $\sin(m/2) \ge 0$ and $2\cos(m/2) \ge 1 \ge \cos((M-b)/2)$. This proves the first sign. If $0 < m < c$, the sine is positive and $2\cos(m/2) > 1$, so the sign is strict.

For the largest gap, the same addition formula gives

$$F_M = 2\sin(M/2)\big[2\cos(M/2) - \cos((b-m)/2)\big]. \tag{14}$$

If $\pi \le M \le 2\pi$, then $0 \le b - m \le b + m = 2\pi - M \le \pi$. Thus $\cos((b-m)/2) \ge 0$ and $\cos(M/2) \le 0$. Also $\sin(M/2) \ge 0$, so (14) is nonpositive.

If $c \le M \le \pi$, the inequality $b \le M$ gives

$$0 \le b - m = 2b - (2\pi - M) \le 3M - 2\pi \le \pi.$$

Put $h = M/2$. Since cosine decreases on $[0, \pi/2]$,

$$\cos((b-m)/2) \ge \cos(3h - \pi) = -\cos(3h).$$

It follows that

$$\begin{aligned} 2\cos h - \cos((b-m)/2) &\le 2\cos h + \cos(3h) \\ &= \cos h\big(4\cos^2 h - 1\big) \le 0, \end{aligned} \tag{15}$$

since $\pi/3 \le h \le \pi/2$ implies $0 \le \cos h \le 1/2$. Together with (14), this proves $F_M \le 0$ in the remaining case. ∎

**Lemma 2.** *If $0 < a < 1/3$, then $T_a$ preserves coordinate order and maps $\Delta$ into itself. Each face of $\Delta$ is invariant. The literal lift $\widetilde{G}_a$ preserves both $A$ and $B$.*

**Proof.** Let $H_a(t) = t + 3a\sin t$. For $0 \le t \le 2\pi$,

$$H_a'(t) = 1 + 3a\cos t \ge 1 - 3a > 0.$$

As

$$(T_a r)_i = H_a(r_i) - a\sum_{j=1}^{3} \sin r_j,$$

all coordinates are transformed by the same strictly increasing function and then shifted by the same number. Their order, including ties, is preserved.

The updated coordinate corresponding to the minimum $m$ is $m + aF_m \ge m \ge 0$, by Lemma 1. It is still minimal, so all updated coordinates are nonnegative. Their sum is $2\pi$ because $\sum_i F_i = 0$. Thus $T_a(\Delta) \subseteq \Delta$.

If a gap is zero, the other two sum to $2\pi$ and their sines cancel. More explicitly, up to permutation,

$$T_a(0, t, 2\pi - t) = (0, t + 3a\sin t, 2\pi - t - 3a\sin t), \qquad 0 \le t \le 2\pi. \tag{16}$$

Hence every face is invariant, including its vertices. Finally, (11) and (12) translate simplex invariance into $0 \le X \le Y \le 2\pi$ on $A$ and $0 \le Y \le X \le 2\pi$ on $B$. ∎

In sorted coordinates $r_1 \le r_2 \le r_3$ and $w = T_a r$, the two lemmas give

$$w_1 \ge r_1, \qquad w_3 \le r_3, \qquad w_2 + w_3 \le r_2 + r_3, \qquad \sum_i w_i = \sum_i r_i.$$

These are the three-coordinate majorization inequalities $w \prec r$ in the usual convention [2]. The preserved order lets us obtain the quantitative remainder needed here by summation by parts.

# 4 The energy loss and its zero set

**Proof of Theorem 1.** By Section 2 and Lemma 2, only the decrement and its zero set remain to be proved.

Fix $r \in \Delta$, set $w = T_a r$ and $d = w - r$, and apply a common permutation so that $r_1 \le r_2 \le r_3$. Then

$$w_1 \le w_2 \le w_3, \qquad d_1 \ge 0, \quad d_3 \le 0, \qquad d_1 + d_2 + d_3 = 0.$$

Eliminating $d_2$ gives the exact identity

$$w \cdot d = (w_1 - w_2)d_1 + (w_3 - w_2)d_3 \le 0. \tag{17}$$

Expanding the quadratic at the updated point, rather than at the old one, now yields

$$\begin{aligned} E(w) - E(r) &= (w - c\mathbf{1}) \cdot d - \frac{1}{2}\|d\|^2 \\ &= w \cdot d - \frac{1}{2}\|d\|^2 \le -\frac{1}{2}\|d\|^2. \end{aligned} \tag{18}$$

This proves (5) for every point of the closed simplex. The energy identities and triangle invariance transfer it to (6).

We also check the meaning of its squared displacement on the quotient. On the diagonal both gap formulas give $(t, 0, 2\pi - t)$. Across a vertical seam, $(0, t)$ and $(2\pi, t)$ have gap triples

$$(0, t, 2\pi - t), \qquad (t, 2\pi - t, 0),$$

respectively. The horizontal seam gives the same pair of triples, and the corners give permutations of $(0, 0, 2\pi)$. Since $T_a$ commutes with coordinate permutations and the Euclidean norm is permutation invariant, $\|T_a r - r\|^2$ agrees at all identified representatives.

By (5), zero energy loss is equivalent to $T_a r = r$, or to $F(r) = 0$ because $a > 0$. Sort the gaps again as $m \le b \le M$. If $m > 0$, Lemma 1 excludes stationarity unless $m = c$. The sum and order then force $m = b = M = c$. If $m = 0$, formula (16) shows that

stationarity requires $\sin b = 0$. Here $0 \le b \le \pi$, so $b = 0$ or $b = \pi$. Thus the complete stationary gap set is

$$\{(c, c, c)\} \ \cup \ \mathrm{Perm}(0, \pi, \pi) \ \cup \ \mathrm{Perm}(0, 0, 2\pi), \tag{19}$$

where Perm denotes all coordinate permutations. Each listed triple is stationary by substitution.

On $A$ the inverse gap map is $(x, y) = (r_1, r_1 + r_2)$; on $B$ it is $(x, y) = (r_1 + r_2, r_1)$. The central triple gives $p_+$ and $p_-$. The permutations of $(0, \pi, \pi)$ give, on $A$,

$$(0, \pi), \quad (\pi, \pi), \quad (\pi, 2\pi) \equiv (\pi, 0),$$

and the lower triangle gives the same three torus points. The permutations of $(0, 0, 2\pi)$ give square corners, all representing $(0, 0)$. These are exactly the six points in (7).

Stationary gaps and the affine inverse formulas show that all six are fixed points of the literal map. Conversely, every torus fixed point has zero energy loss and hence belongs to this list. At every other point, the squared displacement in (6) is positive. The proof is complete. ∎

*Remark.* A complete Lyapunov function in the sense recalled by Norton [3, p. 594] must separate distinct chain-recurrent classes, in addition to satisfying a decrease condition. The two attracting fixed points identified in [1, Theorems 6 and 7] belong to distinct chain classes, whereas (2) assigns value zero to both. Thus this particular function does not satisfy that separation requirement. Theorem 1 verifies its strict-decrease property, not the full completeness wording of Conjecture 1.

The theorem concerns the prescribed quadratic and $0 < a < 1/3$; the basin description is already given in [1]. The accompanying symbolic checker verifies algebraic identities and fixed-point substitutions. All real-variable signs and the exhaustive stationary-point argument are proved above.

# References


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